Middle East Technical University Department of Computer Engineering

CENG 495

Cloud Computing

2025-2026 Spring

Take Home Exam 1

(Ver. 1.0)

April 5 2026, Sunday 23:59

1 Introduction

For the assignment, your task is to develop an e-commerce web application and deploy it to the cloud Platform-as-a-Service (PaaS) Vercel[1]. You will use the NoSQL[2] database MongoDB[3] on the Database-as-a-Service (DBaaS) cloud platform MongoDB Atlas[4].

2 Atlas (MongoDB Platform)

- Create a free account on MongoDB Atlas MongoDB Atlas[5].
- Create a new project and deploy a new database. You can find the documentation here[6].
- Configure your database to be accessible from any IP address (from the `Network Access` Section), and setup password authorization for DB access (from the `Database Access` section).
- Atlas allows for fine-grained database access controls under the `Security` Section. You can leverage this for your admin user and regular users.

3 Vercel (App PaaS)

- Login to Vercel using a new or an existing account.
- Review the documentation[7] and choose the technology stack to implement the assignment. There are no restrictions on which language or stack you can use to implement, but it is recommended to use the language or the stack you are familiar with.
- In the documentation, you may notice that Vercel expects a GitHub repo to pull and deploy your application from. You should use a private repository for that. If you wish, you can make your repository public after the grades are announced.

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[1] https://vercel.com
[2] https://yurichev.org/NoSQL/
[3] https://www.mongodb.com/
[4] https://www.mongodb.com/atlas
[5] https://www.mongodb.com/atlas/database
[6] https://www.mongodb.com/docs/
[7] https://vercel.com/docs

4 E-Commerce Application

You are probably familiar with many examples of e-commerce sites for purchasing products like groceries or clothing. However, if you want inspiration for your design, sites using Shopify[8] platform are common. And if you want inspiration for the architecture for your project, examining this project[9] may be a good starting point.

However, the features expected from your e-commerce application is far more limited in scope compared to a real e-commerce site, and they can be found in the following sections.

4.1 Home Page

This is the start page of your application. It is accessed through `index.html`. The items available for sale should be listed on the homepage with filtering options for the category of the items. For the purposes of this homework, **at minimum**, the following categories of items should be for sale;

- Vinyls
- Antique Furniture
- GPS Sport Watches
- Running Shoes
- Camping Tents

If the current user is not authorized, they are only allowed to browse the items. You should offer a way for user authentication on your home page.

4.2 Admin Capabilities

When someone logs in as the admin, they should be able to do the following:

Add Item: Create and insert a new item into the database. See Section 4.4 for the required attributes of items.

Remove Item: Remove an item from the application. By removing the item, you should also remove the ratings and the reviews relevant to that item. Any user affected by the removal should have their relevant fields updated as well.

Add User: Create a new user. Refer to Section 4.5 for user attributes.

Remove User: Remove the user and update relevant items that are effected by this change.

4.3 Regular User Capabilities

When a regular user logs in, they should be able to do the following on each item's page:

Rate Item: Rate an item on a scale of 1 to 5. The average rating of the item is updated as a result. A user can rate an item multiple times, the latter overwriting the prior rating.

Review Item: Review an item. Reviews can also be updated. When a review is updated, "`Edit:`" string and the new review should be appended to the end of the original review.

[8] https://www.shopify.com/

[9] https://github.com/venkataravuri/e-commerce-microservices-sample

4.4 Item Attributes

The following is the list of attributes for items. When implementing the item details view (either a new page or a frame), ensure that every relevant detail of the item is presented to the user. Note that not every field is required for every item.

Name: The name of the item.

Description: The description of the item.

Price: The price of the item, choice of currency is left to you.

Seller: The seller of the item.

Image: The image showing the item. You do not need to implement image upload, a hyperlink to an image file on the Internet is fine.

Battery Life: Only relevant for GPS sport watches.

Age: Only relevant for antique furniture and vinyls.

Size: Only relevant for the running shoes.

Material: Only relevant for antique furniture and running shoes.

Rating: Average rating this item has received.

Reviews: List of the reviews on the item.

Condition: Whether the item is new or used.

You are free to add additional attributes as you see fit. For instance, to keep track of the average rating, you can also have a **number of reviewers**.

4.5 User Attributes

Create a dedicated page presenting the following details of the currently authenticated regular user. If the current user is not logged-in, they should not be able to see this page or the way of accessing this page.

Username: The username of the user

Average Rating: The average of every rating given by this user

Reviews: Every review written by this user

5 Rules and Submission

- This is an exam. Plagiarism will not be tolerated.
- Upload a **zip** file contains all of your source code to ODTUClass assignment page before the deadline.
  - The penalty for late submission is $(\textbf{Number of Late Days})^{\mathbf{2}} \times \mathbf{5}$.
  - You will get 0 automatically after 3 late days.
- Your submission must include a `README` file that explains your design decisions and the **publicly accessible** URL of your Vercel deployment. It should include explanations on:
  - How to login as a regular user and the admin user and how to use the application

- Rationale behind the choices regarding:
  - The selected programming language and frameworks
  - The general architecture of the application
- And other points as you would like to mention

- Populate your application by adding at least 8 items and 3 users. Have your users rate and review every item at least once. Make sure at least one item from every category is available in the application.

- Since we are using a NoSQL database which allows for dynamic schemas, you are expected to take advantage of this flexibility, for example, by designing your database with the fewest collections possible.

- We may request live demo sessions for randomly selected submissions and submissions we deem necessary. Details regarding how and when the session will take place will be announced later on ODTUClass page.